

Derivation of Maxwell's Equations

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Maxwell's Equations

All the equations we will discuss in this note:

$$\text{Maxwell's Eqs:} \left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = \frac{\varrho}{\epsilon_0} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{array} \right. \quad \text{In Matters:} \left\{ \begin{array}{l} \nabla \cdot \mathbf{D} = \varrho_f \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \end{array} \right.$$

1 Maxwell's Equations

The electric field and Magnetic field are respectively

$$\left\{ \begin{array}{l} \text{Coulomb's Law : } \mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\varrho(\mathbf{r}') \hat{\mathbf{r}}}{r^2} d\tau' \\ \text{Biot-Sarvart's Law : } \mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau'. \end{array} \right. \quad (1)$$

Where $\mathbf{r} = \mathbf{r} - \mathbf{r}'$.

1.1 Gauss's Law

The electric field $\mathbf{E}(\mathbf{r})$ is given by

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\varrho(\mathbf{r}') \hat{\mathbf{r}}}{r^2} d\tau'.$$

Apply the divergence to the field with respect to the *Observation* coordinate($\mathbf{r}$), we have

$$\nabla \cdot \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \varrho(\mathbf{r}') \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) d\tau', \quad (2)$$

with the identity

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi\delta^3(\mathbf{r}). \quad (3)$$

So, (2) becomes:

$$\nabla \cdot \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \varrho(\mathbf{r}') 4\pi\delta^3(\mathbf{r}) d\tau' = \frac{1}{4\pi\epsilon_0} \frac{1}{|-1|} \int \varrho(\mathbf{r}') 4\pi\delta^3(\mathbf{r}' - \mathbf{r}) d\tau', \quad (4)$$

and we can finally obtain

$$\boxed{\nabla \cdot \mathbf{E} = \frac{\varrho(\mathbf{r})}{\epsilon_0}.} \quad (5)$$

The divergence theorem gives

$$\int \nabla \cdot \mathbf{E} d\tau = \oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} \int \varrho(\mathbf{r}) d\tau = \frac{Q_{\text{enc}}}{\epsilon_0},$$

and we thus obtain the integration form

$$\boxed{\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}.} \quad (6)$$

Note that we are discussing how the source in *source* coordinates($\mathbf{r}'$) affects the field itself in the *observation* coordinates($\mathbf{r}$). That is to say, although the charges are distributed over the source coordinates, their collective contributions generate an electric field evaluated at the observation coordinates.

1.2 Magnetic Monopoles

The Biot-Savart law reads

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau'. \quad (7)$$

Then the divergence of $\mathbf{B}(\mathbf{r})$ is:

$$\begin{aligned} \nabla \cdot \mathbf{B}(\mathbf{r}) &= \frac{\mu_0}{4\pi} \int \nabla \cdot \left(\mathbf{J}(\mathbf{r}') \times \frac{\hat{\mathbf{r}}}{r^2} \right) d\tau' \\ &= \frac{\mu_0}{4\pi} \int \left[\underbrace{\frac{\hat{\mathbf{r}}}{r^2} \cdot (\nabla \times \mathbf{J}(\mathbf{r}')) - \mathbf{J}(\mathbf{r}') \cdot \left(\nabla \times \frac{\hat{\mathbf{r}}}{r^2} \right)}_{\text{Product Rules (iii)}} \right] d\tau' \\ &= \frac{\mu_0}{4\pi} \int \left\{ \frac{\hat{\mathbf{r}}}{r^2} \cdot (\nabla \times \mathbf{J}(\mathbf{r}')) + \mathbf{J}(\mathbf{r}') \cdot \left[\nabla \times \left(\frac{\nabla}{r} \right) \right] \right\} d\tau'. \end{aligned} \quad (8)$$

Our focus is on the properties of the field at the observation coordinates. Because the current density is defined in the source coordinates, differential operators with respect to the observation coordinates do not act on the current density and thus yield no contribution. That is, the volume current density $\mathbf{J}(\mathbf{r}')$ is the function of $\mathbf{r}' = (x', y', z')$, so $\nabla \times \mathbf{J}(\mathbf{r}') = 0$, which gives that

$$\boxed{\nabla \cdot \mathbf{B}(\mathbf{r}) = 0.} \quad (9)$$

This equation implies that **there are no magnetic monopoles**.

1.3 Faraday's Law

Φ is the magnetic flux, which is defined as

$$\Phi = \int \mathbf{B} \cdot d\mathbf{a}.$$

Within an infinitesimal time difference dt , the surface placed in the magnetic field $\mathbf{B}$ moves with velocity $\mathbf{v}$ to another position without changing its shape. There exists a 'ribbon' connecting the new position of the surface to the old one. In this case, the infinitesimal area(parallelogram) on the ribbon $d\mathbf{a}$ can be denoted by

$$d\mathbf{a} = (\mathbf{v}dt) \times d\boldsymbol{\ell},$$

where $d\boldsymbol{\ell}$ is one of the sides of the infinitesimal displacement of the parallelogram. Then the flux difference is

$$\begin{aligned} d\Phi &= \int_{\text{ribbon}} \mathbf{B} \cdot d\mathbf{a} = \oint \mathbf{B} \cdot [(\mathbf{v}dt) \times d\boldsymbol{\ell}] \\ \Rightarrow \frac{d\Phi}{dt} &= \oint \mathbf{B} \cdot (\mathbf{v} \times d\boldsymbol{\ell}) = - \underbrace{\oint (\mathbf{v} \times \mathbf{B}) \cdot d\boldsymbol{\ell}}_{\text{Triple Products (iii)}} = - \oint \mathbf{f} \cdot d\boldsymbol{\ell}, \end{aligned}$$

where $\mathbf{f} := \mathbf{v} \times \mathbf{B}$. The **Flux Rule** for motional emf gives that

$$\mathcal{E} = -\frac{d\Phi}{dt}, \quad (10)$$

The emf is defined as

$$\mathcal{E} := \oint \mathbf{f} \cdot d\boldsymbol{\ell}, \quad (11)$$

where $\mathbf{f} = \mathbf{f}_s + \mathbf{E}$, which represents the *force per unit charge*. The meaning of emf is the accumulation of the effect caused by the *source*($\mathbf{f}_s$) confined on the circuits or systems

and the induced electric field $\mathbf{E}$.¹ We consider that there's only the induced electric field that accounts for the emf, that is, with the flux rule

$$\mathcal{E} = \oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt}, \quad (12)$$

Apply the Stokes' theorem, (12) becomes

$$\begin{aligned} \mathcal{E} &= \oint \mathbf{E} \cdot d\boldsymbol{\ell} = \int (\boldsymbol{\nabla} \times \mathbf{E}) \cdot d\mathbf{a} = -\frac{d\Phi}{dt} \\ &= -\frac{d}{dt} \int \mathbf{B}(\mathbf{r}, t) \cdot d\mathbf{a} = \int -\frac{\partial \mathbf{B}(\mathbf{r}, t)}{\partial t} \cdot d\mathbf{a}. \end{aligned}$$

Therefore, we have

$$\boxed{\boldsymbol{\nabla} \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{B}(\mathbf{r}, t)}{\partial t}}. \quad (13)$$

This equation implies that **a changing magnetic field induces an electric field.**

1.4 Ampère's Law with Maxwell's Correction

We apply the curl operator on the Biot-Savart's field:

$$\begin{aligned} \boldsymbol{\nabla} \times \mathbf{B} &= \frac{\mu_0}{4\pi} \int \boldsymbol{\nabla} \times \left(\mathbf{J}(\mathbf{r}', t) \times \frac{\hat{\boldsymbol{\mathbf{r}}}}{r^2} \right) d\tau' \\ &= \frac{\mu_0}{4\pi} \int \left[\underbrace{\left(\frac{\hat{\boldsymbol{\mathbf{r}}}}{r^2} \cdot \boldsymbol{\nabla} \right) \mathbf{J}(\mathbf{r}', t) - \left(\mathbf{J}(\mathbf{r}', t) \cdot \boldsymbol{\nabla} \right) \frac{\hat{\boldsymbol{\mathbf{r}}}}{r^2} + \mathbf{J} \left(\boldsymbol{\nabla} \cdot \frac{\hat{\boldsymbol{\mathbf{r}}}}{r^2} \right) - \frac{\hat{\boldsymbol{\mathbf{r}}}}{r^2} \left(\boldsymbol{\nabla} \cdot \mathbf{J}(\mathbf{r}', t) \right)}_{\text{Product Rules (vi)}} \right] d\tau' \end{aligned}$$

This can be reduced to

$$\boldsymbol{\nabla} \times \mathbf{B} = \frac{\mu_0}{4\pi} \int \left[\mathbf{J}(\mathbf{r}', t) \left(\boldsymbol{\nabla} \cdot \frac{\hat{\boldsymbol{\mathbf{r}}}}{r^2} \right) - \left(\mathbf{J}(\mathbf{r}', t) \cdot \boldsymbol{\nabla} \right) \frac{\hat{\boldsymbol{\mathbf{r}}}}{r^2} \right] d\tau'. \quad (14)$$

By the delta function identities (i), we have

$$\mathbf{J}(\mathbf{r}', t) \left(\boldsymbol{\nabla} \cdot \frac{\hat{\boldsymbol{\mathbf{r}}}}{r^2} \right) = \mathbf{J}(\mathbf{r}', t) 4\pi \delta^3(\mathbf{r}). \quad (15)$$

And by the delta function identities (ii) & (iii), the linear operators give that

$$-\boldsymbol{\nabla}_{\mathbf{r}} = \boldsymbol{\nabla}'_{\mathbf{r}'},$$

¹When the electric field is electrostatic, then $\mathcal{E} = \oint \mathbf{E} \cdot d\boldsymbol{\ell} = 0$, which means that there's only the source $\mathbf{f}_s$ can cause the emf.

then (14) becomes

$$\begin{aligned}\nabla \times \mathbf{B} &= \frac{\mu_0}{4\pi} \int \left[\mathbf{J}(\mathbf{r}', t) 4\pi \delta^3(\mathbf{r}) + \left(\mathbf{J}(\mathbf{r}', t) \cdot \nabla' \right) \frac{\hat{\mathbf{r}}}{r^2} \right] d\tau' \\ &= \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}', t) 4\pi \delta^3(\mathbf{r}) d\tau' = \mu_0 \mathbf{J}(\mathbf{r}, t).\end{aligned}\tag{16}$$

Note that the second term of the integral is

$$\begin{aligned}I_i &= \int_V \left[\left(\mathbf{J} \cdot \nabla' \right) \frac{\hat{\mathbf{r}}}{r^2} \right]_i d\tau' = \int_V \left(\mathbf{J} \cdot \nabla' \right) \frac{r_i}{r^3} d\tau' \\ &= \int_V \nabla' \cdot \left(\frac{r_i}{r^3} \mathbf{J} \right) d\tau' - \int_V \frac{r_i}{r^3} \left(\underbrace{\nabla' \cdot \mathbf{J}(\mathbf{r}', t)}_{\text{Charge Conservation}} \right) d\tau' \\ &= \oint_{\partial V} \frac{r_i}{r^3} (\mathbf{J} \cdot d\mathbf{a}') \quad (\text{Divergence Theorem}) \\ &= 0 \quad (\forall i)\end{aligned}$$

What's crucial is that the current on the boundary is **zero**, which means that we consider the current safely flows inside the volume in the space we discuss. The continuity equation of the charge and Gauss's law indicate that

$$\nabla \cdot \mathbf{J}(\mathbf{r}, t) = -\frac{\partial \varrho(\mathbf{r}, t)}{\partial t} = -\frac{\partial}{\partial t} \left[\nabla \cdot (\epsilon_0 \mathbf{E}(\mathbf{r}, t)) \right] = -\nabla \cdot \left(\epsilon_0 \frac{\partial \mathbf{E}(\mathbf{r}, t)}{\partial t} \right), \tag{17}$$

(17) can be written as

$$\nabla \cdot \left[\mathbf{J}(\mathbf{r}, t) + \epsilon_0 \frac{\partial \mathbf{E}(\mathbf{r}, t)}{\partial t} \right] = 0 \tag{18}$$

Substituting this into (16) to make the dimension correct, we have

$$\nabla \cdot (\nabla \times \mathbf{B}) = \nabla \cdot \left[\mu_0 \mathbf{J}(\mathbf{r}, t) + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}(\mathbf{r}, t)}{\partial t} \right] = 0 \tag{19}$$

Therefore, we obtain the final equation, which is called *Ampère's Law* fixed by Maxwell:

$$\boxed{\nabla \times \mathbf{B} = \mu_0 \mathbf{J}(\mathbf{r}, t) + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}(\mathbf{r}, t)}{\partial t}} \tag{20}$$

2 Maxwell's Equations in Matters

2.1 Gauss's Law with Electric Displacement

In the polarized source, the bound charge density is given by the dipole moment in the electric field:

$$\varrho_b = -\nabla \cdot \mathbf{P}. \quad (21)$$

Gauss's law states that

$$\nabla \cdot \mathbf{E} = \frac{\varrho}{\epsilon_0} = \frac{\varrho_f + \varrho_b}{\epsilon_0} = \frac{\varrho_f - \nabla \cdot \mathbf{P}}{\epsilon_0},$$

this can be simplified as

$$\nabla \cdot (\epsilon_0 \mathbf{E}) + \nabla \cdot \mathbf{P} = \nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \varrho_f. \quad (22)$$

We define the electric displacement $\mathbf{D}$ as²

$$\mathbf{D} := \epsilon_0 \mathbf{E} + \mathbf{P}, \quad (23)$$

then Gauss's law in the dielectric is corrected to

$$\boxed{\nabla \cdot \mathbf{D} = \varrho_f}. \quad (24)$$

2.2 Ampère's Law with Auxiliary Field

The Magnetization $\mathbf{M}$ results in a bound current density

$$\mathbf{J}_b = \nabla \times \mathbf{M}. \quad (25)$$

However, in the electric field, the dielectric may be polarized. During the process of polarization, the bound charge density σ_b varies in time, which yields the current called *polarization current*. If $|\mathbf{P}|$ increases, the charge on each end increases accordingly, which yields

$$dI = \frac{d}{dt}(\sigma_b da_\perp) = \frac{\partial \sigma_b}{\partial t} da_\perp = \frac{\partial \mathbf{P}}{\partial t} \cdot \hat{\mathbf{n}} da = \frac{\partial P}{\partial t} da_\perp \quad (26)$$

This can be written as

$$\frac{d\mathbf{I}}{da_\perp} \cdot \hat{\mathbf{n}} = \mathbf{J}_p \cdot \hat{\mathbf{n}} = \frac{\partial \mathbf{P}}{\partial t} \cdot \hat{\mathbf{n}} \Rightarrow \mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t}.$$

This means that the varied electric field may affect the polarization and further induce the current by the varied dipole moment. Then the current density can be separated into three parts:³

$$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p = \mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} \quad (27)$$

²Recall: The electric displacement of a **linear dielectric** is $\mathbf{D} = \epsilon_0(1 + \chi_e)\mathbf{E} = \epsilon_0\epsilon_r\mathbf{E} = \epsilon\mathbf{E}$.

³In the electrostatic field, we can only separate it into $\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b$.

Substituting this result into Ampère's law, we have

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \left(\mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} \right) + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Dividing both sides by μ_0 , we can obtain

$$\begin{aligned} \frac{1}{\mu_0} \nabla \times \mathbf{B} &= \mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \\ \Rightarrow \nabla \times \left(\frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \right) &= \mathbf{J}_f + \frac{\partial}{\partial t} \left(\underbrace{\epsilon_0 \mathbf{E} + \mathbf{P}}_{\mathbf{D}} \right). \end{aligned}$$

The auxiliary field $\mathbf{H}$ is defined as

$$\mathbf{H} := \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}, \quad (28)$$

so, Ampère's Law in matters is affected by the varied magnetic field(magnetization) and the varied electric field(polarization), it's denoted by

$$\boxed{\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}}. \quad (29)$$

Some Identities

Triple Products:

$$\begin{cases} \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \\ \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \quad \text{Bac(k)-Cab law} \end{cases}$$

Products Rules:

$$\begin{cases} \nabla(fg) = f(\nabla g) + g(\nabla f) \\ \nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A} \\ \nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) \\ \nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot \nabla f \\ \nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times \nabla f \\ \nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) \end{cases}$$

Second Derivatives:

$$\begin{cases} \nabla \cdot (\nabla \times \mathbf{A}) = 0 \\ \nabla \times (\nabla f) = \mathbf{0} \\ \nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \end{cases}$$

Green's Identities and Some Integrals:

$$\left\{ \begin{array}{l} \int_{\mathcal{V}} (T \nabla^2 U - U \nabla^2 T) d\tau = \oint_S (T \nabla U - U \nabla T) \cdot d\mathbf{a} \\ \int_S \nabla T \times d\mathbf{a} = - \oint_{\mathcal{P}} T d\boldsymbol{\ell} \\ \oint (\mathbf{c} \cdot \mathbf{r}) d\boldsymbol{\ell} = \left(\int d\mathbf{a} \right) \times \mathbf{c} \\ \int_{\mathcal{V}} \nabla \times \mathbf{F} d\tau = - \oint_S \mathbf{F} \times d\mathbf{a} \end{array} \right.$$

Delta functions:

$$\left\{ \begin{array}{l} \nabla \cdot \left(\frac{\hat{\boldsymbol{r}}}{r^2} \right) = 4\pi\delta^3(\boldsymbol{r}) \\ \nabla \left(\frac{1}{r} \right) = \nabla \left(\frac{1}{\underbrace{\mathbf{r} - \mathbf{r}'}_{\text{distance}}} \right) = -\frac{\hat{\boldsymbol{r}}}{r^2} \\ \nabla' \left(\frac{1}{r} \right) = \nabla' \left(\frac{1}{\underbrace{\mathbf{r} - \mathbf{r}'}_{\text{distance}}} \right) = \frac{\hat{\boldsymbol{r}}}{r^2} \\ \nabla^2 \left(\frac{1}{r} \right) = -4\pi\delta^3(\boldsymbol{r}) \end{array} \right.$$